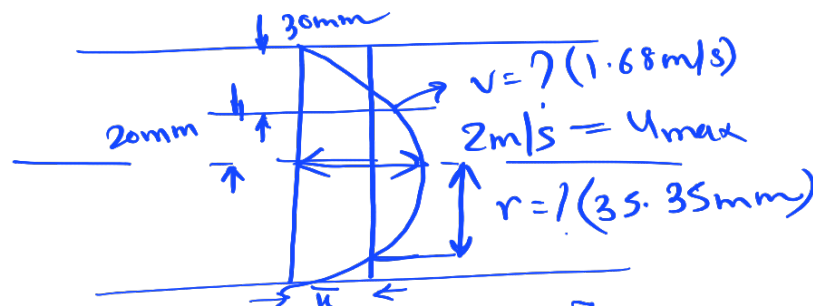


A crude oil of viscosity 0.9 poise and sp. gr. 0.8 is flowing through a horizontal circular pipe of diameter 80 mm and of length 15 m. Calculate the difference of pressure at the two ends of the pipe, if 50 kg of the oil is collected in a tank in 15 seconds. [Ans. 0.559 N/cm²]

$$\begin{aligned}
 Re &= \frac{\rho v D}{\mu} \\
 &= 800 \times \left(\frac{\frac{50}{15}}{\frac{800}{800}} \right) \times \frac{80}{1000} \\
 &\quad \frac{\pi/4 \times \left(\frac{80}{1000} \right)^2}{0.09} \\
 &= 589 < 2000 \rightarrow \text{Laminar flow /} \\
 &\quad \text{viscom flow}
 \end{aligned}$$

$$\begin{aligned}
 P_1 - P_2 &= \frac{32 \mu \bar{u} L}{D^2} = \frac{32 \times 0.09 \times 0.829 \times 15}{\left(\frac{80}{1000} \right)^2} \\
 &= 5595 \text{ N/m}^2
 \end{aligned}$$

A viscous flow is taking place in a pipe of diameter 100 mm. The maximum velocity is 2 m/s. Find the mean velocity and the radius at which this occurs. Also calculate the velocity at 30 mm from the wall of the pipe.
 [Ans. 1 m/s, $r = 35.35$ mm, $u = 1.68$ m/s]



$$u = -\frac{1}{4\mu} \frac{\partial P}{\partial x} [R^2 - r^2]$$

$$u = u_{max} \text{ at } r = 0 \quad R = \frac{50}{1000}$$

$$\Rightarrow 2 = -\frac{1}{4\mu} \frac{\partial P}{\partial x} [R^2 - 0]$$

$$\Rightarrow -\frac{1}{4\mu} \frac{\partial P}{\partial x} = 800 \quad \frac{1}{\text{ms}}$$

$$u = u_{avg} \text{ at } r = ?$$

$$1 = \underbrace{-\frac{1}{4\mu} \frac{\partial P}{\partial x}}_{800} \left[\left(\frac{50}{1000} \right)^2 - r^2 \right]$$

$$\Rightarrow r = 35.35 \text{ mm}$$

$$u \text{ at } r = 20 \text{ mm}$$

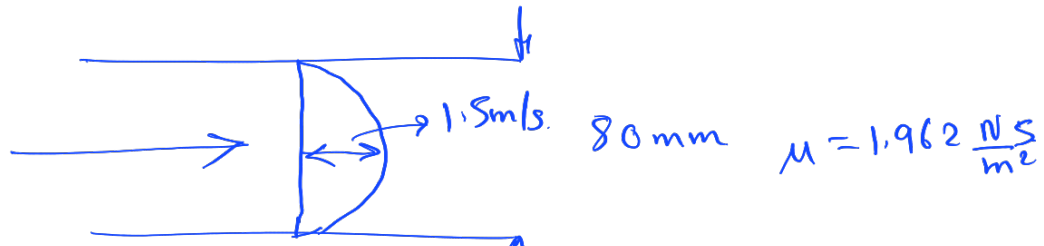
$$= -\frac{1}{4\mu} \frac{\partial P}{\partial x} \left[\left(\frac{50}{1000} \right)^2 - \left(\frac{20}{1000} \right)^2 \right]$$

$$= 800 \times \left[\left(\frac{50}{1000} \right)^2 - \left(\frac{20}{1000} \right)^2 \right]$$

$$= 1.68 \text{ m/s.}$$

Determine (a) the pressure gradient, (b) the shear stress at the two horizontal parallel plates and (c) the discharge per metre width for the laminar flow of oil with a maximum velocity of 1.5 m/s between two horizontal parallel fixed plates which are 80 mm apart. Take viscosity of oil as $\frac{1.962 \text{ Ns}}{\text{m}^2}$.

[Ans. (a) - 3678.7 N/m² per m, (b) 147.15 N/m², (c) .08 m³/s]



$$\frac{\partial p}{\partial x} \quad u = -\frac{1}{2\mu} \frac{\partial p}{\partial x} [ty - y^2]$$

$$u = u_{\max} \text{ at } y = \frac{t}{2}$$

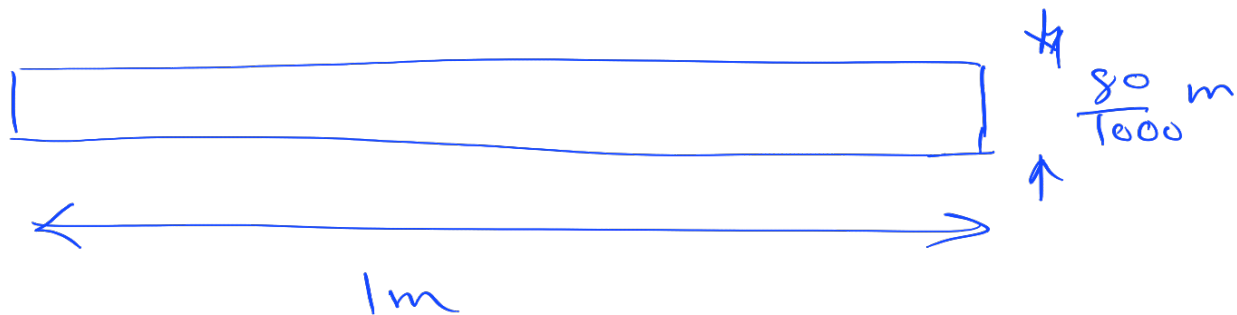
$$\Rightarrow 1.5 = -\frac{1}{2 \times 1.962} \frac{\partial p}{\partial x} \left[\frac{80}{1000} \times \frac{40}{1000} - \left(\frac{40}{1000} \right)^2 \right]$$

$$\Rightarrow \frac{\partial p}{\partial x} = -3678.75 \text{ N/m}^2 / \text{m}$$

$$\tau = -\frac{1}{2} \frac{\partial p}{\partial x} [t - 2y]$$

$$= -\frac{1}{2} (-3678.75) \times \left(\frac{80}{1000} - 2 \times 0 \right)$$

$$= 147.15 \text{ N/m}^2$$



$$Q = 0.08 \text{ m}^3/\text{s}$$

Water is flowing between two large parallel plates which are 2.0 mm apart. Determine : (a) maximum velocity, (b) the pressure drop per unit length and (c) the shear stress at walls of the plate if the average velocity is 0.4 m/s. Take viscosity of water as 0.01 poise.

[Ans. (a) 0.6 m/s, (b) 1199.7 N/m² per m, (c) 1.199 N/m²]

$$u_{max} = u_{avg} \times \frac{3}{2} = 0.6 \text{ m/s}$$

$$u = -\frac{1}{2\mu} \frac{\partial P}{\partial x} [ty - y^2]$$

$$u = u_{max} \text{ at } y = t/2$$

$$\Rightarrow \frac{\partial P}{\partial x} = -1200 \text{ N/m}^2 / \text{m}$$

$$\tau = -\frac{1}{2} \frac{\partial P}{\partial x} [t - 2y] = -\frac{1}{2} \times (-1200) \left(\frac{2}{1000} - 0 \right) = 1.2 \text{ N/m}^2$$

There is a horizontal crack 50 mm wide and 3 mm deep in a wall of thickness 150 mm. Water leaks through the crack. Find the rate of leakage of water through the crack if the difference of pressure between the two ends of the crack is 245.25 N/m^2 . Take the viscosity of water as 0.01 poise. [Ans. $183.9 \text{ cm}^3/\text{s}$]

$$u = -\frac{1}{2\mu} \frac{\partial P}{\partial x} [ty - y^2]$$

$$u = u_{\max} \text{ at } y = \frac{t}{2}$$

$$u_{\max} = -\frac{1}{2 \times 0.001} \times \frac{245.25}{0.15} \times \left[\frac{3}{1000} \times \frac{1.5}{1000} - \left(\frac{1.5}{1000} \right)^2 \right]$$

$$= 1.839 \text{ m/s.}$$

$$u_{\text{avg}} = \frac{1.839}{1.5} = 1.22 \text{ m/s.}$$

$$Q = u_{\text{avg}} \times \frac{50}{1000} \times \frac{3}{1000} = 0.0001839 \text{ m}^3/\text{s}$$

$$= 183.9 \text{ cm}^3/\text{s.}$$

